4. Quanta

The two pillars of twentieth-century physics – general relativity and quantum mechanics – could not be more different from each other. General relativity is a compact jewel: conceived by a single mind, based on combining previous theories, it is a simple and coherent vision of gravity, space and time. Quantum mechanics, or quantum theory, on the other hand, emerges from experiments in the course of a long gestation over a quarter of a century, to which many have contributed; achieves unequalled experimental success and leads to applications which have transformed our everyday lives (the computer on which I write, for instance); but, more than a century after its birth, it remains shrouded in obscurity and incomprehensibility.

This chapter illustrates the strange physics of this theory, relates how the theory came into being and the three aspects of reality it has unveiled: granularity, indeterminism and relationality.

Albert again

It's said that quantum mechanics was born precisely in 1900, virtually ushering in a century of intense thought. In 1900 the German physicist Max Planck tries to compute the amount of electromagnetic waves in equilibrium in a hot box. To obtain a formula reproducing the experimental results, he ends up using a trick which does not appear to make much sense: he assumes that the energy of the electric field is distributed in 'quanta', that is to say, in small packets, little bricks of energy. The size of the packets, he assumes, depends on the frequency (that is, the colour) of the electromagnetic waves. For waves of frequency ν, every quantum, or every packet, has energy

$$E = h\nu$$

This formula is the first of quantum mechanics; h is a novel constant which today we call the Planck constant. It fixes how much energy there is in each packet of energy, for radiation of frequency (colour) ν. The constant h determines the scale of all quantum phenomena.

The idea that energy could be made up of finite packets is at odds with everything that was known at the time: energy was considered something that